

Swaroop Mudholkar

Electronics & Telecommunication Engineering Student | Embedded Systems | Automotive Electronics | GenAI

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🔗 swaroop-mudholkar-portfolio.vercel.app 🐙 github.com/swaroop6031



EDUCATION

B.Tech in Electronics & Telecommunication 2023 – 2027 | Pune
Pimpri Chinchwad College of Engineering

12th Science, Maharashtra Board 2023 | Pune
S. B. Patil College of Science and Commerce

10th, CBSE 2021 | Pune
S. B. Patil Public School

SKILLS

- **Programming & Scripting:** Embedded C, C, C++, Python, CAPL, HTML/CSS, JavaScript
- **Embedded Systems, Automotive & AI:** STM32, ESP32, CAN Bus, Vector CANoe, CANdb+++, Diagnostic Protocols (DTC), Sensor Interfacing (UART, I2C), YOLOv7/v8, OpenCV
- **Developer & Hardware Tools:** STM32CubeIDE, E-Plan Electric P8, MATLAB, Git, Multisim, ST-LINK

CERTIFICATIONS

- **NPTEL (IIT Kharagpur):** Introduction to Machine Learning (*Elite Certificate - 81%*) Jul - Sep 2026
- **Microcontroller Embedded C Programming: Absolute Beginners:** Udemy (FastBit Embedded Brain Academy) Aug 2026
- **MathWorks MATLAB Certifications:** MATLAB Onramp, Simulink Onramp, Simulink Fundamentals, Stateflow Onramp, Machine Learning Onramp, Introduction to Motor Control

PROFESSIONAL EXPERIENCE

Cotmac Electronics Pvt. Ltd., Pune Jun 2025 – Jul 2025
PLC Panel Design and Testing Intern

- Engineered PLC panel schematics in E-Plan Electric P8, ensuring 100% wiring accuracy for industrial automation.
- Executed hardware-in-the-loop (HIL) signal testing and fault diagnostics across industrial control panels.

PCCOE EnTC Internship Cell, Pune Aug 2025 – Aug 2026
Documentation Team Lead

- Directed a student team managing technical documentation, reporting, and candidate tracking for 100+ students.
- Coordinated with departmental faculty and industry leads to streamline internship drives and placement workflows.

Technical Competitions

- Participated in Smart India Hackathon (SIH) 2024, IETE National Level Project Competition, and E-Yantra 2025.

PROJECTS

Proactive Detection of Aspergillus Growth Through Indoor Air Monitoring

Tech: STM32, UART, DHT22, MQ135, LCD

- Architected an STM32-based air-quality monitoring system using UART communication and multi-sensor telemetry to process environmental data streams in real time.
- Authored and presented peer-reviewed research on early fungal spore detection, published in the Scitepress International ICITSC Conference proceedings (2025).

Smart Chair System for Posture Monitoring & Wellness Reminder

Tech: ESP32, Force Resistor Sensors, Ultrasonic and Touch Sensors

- Built an ESP32 edge-monitoring system using multi-sensor fusion to process posture data and trigger feedback.
- Calibrated force and ultrasonic sensor thresholds to accurately distinguish between proper seating and poor posture in real time.

Engine ECU Simulation & CAN Network Validation

Tech: Vector CANoe, CAPL, CANdb++

- Simulated high-speed CAN networks in Vector CANoe with custom .dbc databases for multi-ECU validation.
- Developed CAPL scripts to automate Diagnostic Trouble Code (DTC) fault injection and live signal analysis.

Real-Time Object Detection using YOLOv7 & YOLOv8

Tech: Python, CNN, OpenCV, PyTorch

- Benchmarked YOLOv7 vs YOLOv8 models on PyTorch to optimize inference speed and accuracy for edge devices.
- Processed and annotated custom datasets, evaluating mAP, FPS, and detection accuracy across real-time video feeds.